

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

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TITLE OF PROPOSED PROJECT POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management					SHOW LETTER OF INTENT ID IF APPLICABLE
REQUESTED AMOUNT \$ 296,873	PROPOSED DURATION (1-60 MONTHS) 12 months	REQUESTED STARTING DATE 04/01/2026		SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE	
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By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of the individual's knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this proposal. Further, the proposer is hereby providing certifications regarding conflict of interest, flood hazard insurance, responsible and ethical conduct of research, organizational support, and safe and inclusive working environments for off-campus or off-site research, as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section § 1001).

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By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

Certification Requirement Specified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1))

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that each individual employed by the organization and identified on the proposal as senior/key personnel has been made aware of the certification requirements identified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1)).

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Overview:

Ever since the Industrial Revolution, the predominant paradigm for manufacturing has been based on 'economies of scale' through repeatability, mass production and centralized processes. However, in the Internet age, new paradigms of production have emerged that 'democratize' and distribute the manufacturing of even complex products. Digital fabrication, online collaborative design and now AI-supported technologies are enablers of new production approaches that are based on 'economies of scope' where lots of different products, rather than the same product over-and-over, can be made profitably.

However, quality management has not yet adapted to this new digital, democratized, distributed reality. And now the biggest barrier to the adoption of distributed manufacturing is the challenge of distributed quality management. No open, shared, trusted, interoperable ecosystem currently exists to support democratized quality control and distributed quality assurance, verification, liability and continuous improvement across heterogeneous actors. Indeed, regulatory and certification frameworks have evolved to become significant and costly impediments even in traditional manufacturing, so innovation is needed.

Some disparate open-source products and approaches are emerging for distributed quality management. Examples were shared in the recent 'Distributed Quality Management' gatherings, such as the Global Distributed Tracking (GDT) tool from the Global Open-Source Quality Assurance System (GOSQAS). A community is forming from across open-source world, particularly from the Open Hardware movement.

This Phase 1 project will scope Open Distributed Quality Management (OpenDQM). It will scope an Open-Source Ecosystem (OSE) that operates community-governed infrastructures for stakeholders to collaboratively define, validate and steward quality practices specifically designed for distributed manufacturing environments. The project will map stakeholders and their needs, assess the market, report on the state of the art, engage regulatory and certification bodies to understand their challenges, prototype governance and data-exchange models and assess readiness for a Phase II implementation. It will use training on GDT to explore priorities and issues in this space – particularly with consumers and makers.

Intellectual Merits:

The project advances the science of distributed systems, quality management and socio-technical governance. It will develop the first 'state of the art' insight into diverse existing products and approaches and, by induction from these, develop a new typology for quality management. This will offer new architectures for federated quality assurance where many independent producers can share quality standards, testing protocols and verification data without needing costly centralized control. The project will contribute governance innovation for open technical ecosystems, likely more focused on stewardship than compliance. And the project will advance research into distributed manufacturing and resilient supply chains, particularly research into widespread PPE production in the COVID-19 response.

Broader Impacts:

The scope of the challenge in distributed quality management is enormous – just as it is in other areas that already have diverse paradigms of production, such as food (and, increasingly, news, music, movies and other industries affected by the Internet). This project will focus down that scope and its outputs will reach and impact: the Open Hardware movement; local and distributed manufacturers, and all the users and consumers who benefit from them, and; the quality management sector – particularly government regulators who struggle to stay ahead. It may also impact traditional manufacturers with new, open innovations in quality systems. Overall, massive numbers of small manufacturers could be positively impacted if the challenge of distributed quality management can be met, which would support new high-value jobs everywhere. And of course, it will ensure consumer, public and environmental safety in the new paradigm of production, where almost anyone can make almost anything almost anywhere.

Keywords: ENG; Quality Control; Quality Assurance; Safety; Manufacturing.

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Project Description

1. Introduction

Ever since the Industrial Revolution, manufacturing has relied on ‘economies of scale’ based on repeatability, mass production, and centralized processes. However, new production paradigms have emerged that distribute the manufacturing process, thus enabling the local production of complex products [1]. Digital fabrication, online collaborative design, and open-source designs enable new production approaches that are based on ‘economies of scope’. This new production environment allows manufacturers to make many various products, rather than mass producing the same product, while remaining profitable [2]. Today, anyone can access open-source designs and documentation to make their own products, from children’s toys [3] and assistive devices, [4] to medical diagnostic implements [5] and search and rescue equipment [6]. In 2020, individual makers and small to medium-scale manufacturers responding to the COVID-19 pandemic deployed 3D printers and other additive manufacturing technology to make approximately 48 million items of PPE and other medical devices [7]. By sharing and iterating their design files, they succeeded where traditional supply chains failed [8,9].

However, it is yet unclear how to implement scaled quality management in today’s digital, democratized, distributed reality [1]. While standards and regulatory infrastructure exist for commercial-scale, centralized manufacturing, such standards are poorly matched to the variety and specialization of distributed manufacturing across many, independent manufacturers. Due to the novelty of distributed manufacturing capabilities and the various range of supported products, discussions around quality management in distributed manufacturing have only recently begun. This situation reveals a lack of shared definitions and agreed-upon practices that would be applicable across domains.

Distributed quality management focuses on accurate measurement and transparent reporting of core evaluation metrics across individual instances of a product [1]. Through discussions with key collaborators and designers of hardware produced worldwide, including the Open Flexure Microscope project [10] and the Open Science Hardware Foundation [11], Public Invention has identified that the biggest barrier to the adoption of distributed manufacturing is the challenge of distributed quality management (DQM). The lack of DQM is a key barrier to the adoption of real, well-developed distributed manufacturing across large networks of local makers and manufacturers in both high-resource and low-resource settings.

To solve this disconnect, a Public Invention initiative known as the Global Open-Source Quality Assurance System (GOSQAS) [12] created Global Distributed Tracking (GDT). GDT is a free, fully encrypted, and open-source software enabling closed-loop tracking for products, information, and logistics. Deployed via a mobile-phone friendly website, GDT is currently used by Doctors Without Borders’ “3D Printing for All” project [13], Distributed Medical Device Manufacturing (DMDM) [14], and the Canadian medical solidarity company Glia [5]. While GDT provides a critical foundation for provenance, traceability, and anti-counterfeiting, there are significant dimensions to the challenge of DQM that it cannot address. These include standardization of performance metrics and test protocols comparable across sites and allowable substitutions, such as is administered by the International Organization for Standardization and the American Society for Testing and Materials, open source documentation quality and completeness such as is certified by the Open Source Hardware Association, and transparent and distributed accountability mechanisms such as are supported by GitHub and other collaborative development platforms.

Furthermore, the landscape of distributed manufacturing is changing rapidly as new advances in additive technology and emerging designs like the Open Flexure Microscope gain traction in new fields.

There remains a need for an ecosystem capable of identifying and responding to the full range of DQM challenges – to provide the means of implementing DQM. No open, shared, trusted, interoperable ecosystem currently exists to support distributed quality control and assurance, verification, liability, and continuous improvement across heterogeneous actors. The vision guiding this project is an Open-Source Ecosystem (OSE) for Distributed Quality Management (DQM) - called OpenDQM. This proposed OSE will gather standards and compliance experts, consumers of distributed manufactured hardware, and the designers and maintainers of hardware for distributed manufacturing among other stakeholders to identify and address the frontiers of DQM. OpenDQM will support the development of a geographically distributed, trustworthy manufacturing base of a large number of small, agile firms. This will enable the public to have greater access to safe and effective products, even while these new, distributed paradigms of production become mainstream. These products will be trustworthy because experts and end-users alike will be able to easily evaluate the quality assurance of each product through manufacturing process stages. In this system, makers will be able to access quality standards, tools and systems, while users and consumers will know the product origin and who is responsible for it.

2. Context of OSE

OpenDQM seeks to strengthen and support a culture of quality and accountability in the distributed manufacturing, open hardware, maker and local manufacturing movements. This motivating factor has already inspired new open-source products and approaches, like GDT [12], that are disparately emerging from these movements. Some examples were shared in the recent virtual ‘Distributed Quality Management’ workshops [21] held by Public Invention, including solutions spanning: improving documentation; provenance; tracking; standards; test equipment; localised testing systems; qualification; verification; rating; reviews; policy; governance; insurance; warranties, and; consumer awareness. Ecosystemic action is now needed, particularly regarding those challenges and costs that today must be internalized by each and every maker/manufacture, making all of them less profitable or even less viable, such as governance of liabilities and warranties that enhance consumer safety and confidence.

2.1 Product

This project seeks to establish OpenDQM, an open-source Distributed Quality Management (DQM) ecosystem that can respond to the wide variety of quality challenges arising in distributed manufacturing. OpenDQM will unify dialogue across perspectives and establish shared norms and best practices. The Phase I effort will involve uniting stakeholders to form a coordinated community focused on identifying and evaluating domain-appropriate DQM approaches. Distributed manufacturing encompasses a broad spectrum of products, risk profiles, and regulatory environments. As a result, no single method or tool can adequately address all DQM needs. OpenDQM will convene stakeholders including practitioners, tool developers, regulators, certifiers, insurers, and end users to discuss DQM approaches, procedural norms, and mechanisms.

OpenDQM will examine how existing solutions, such as Global Distributed Tracking (GDT), affect DQM challenges in practice, upon what complementary practices they depend, and how they may combine with other approaches to support end-to-end quality assurance. Through structured engagement, documentation, and synthesis, the OpenDQM ecosystem will produce a curated suite of quality management approaches aligned with a variety of distributed manufacturing contexts. This suite may include technical tools, shared testing and calibration practices, documentation conventions, peer review processes, and third-party verification models.

2.2 Vision and Guiding Principles of the Proposed OSE

The OpenDQM OSE seeks to enable safe and trustworthy distributed manufacturing by supporting the shared and open development of quality practices suited to this new paradigm of production. This approach allows a variety of actors to assess product quality without relying solely on centralized manufacturers, proprietary systems, or brand reputation. The vision for OpenDQM arose from the growing use of open-source designs, digital fabrication, and small-batch production across commercial manufacturing, scientific instruments, healthcare and humanitarian response. These areas heavily rely on quality assurance to ensure that each product is safe to use. However, existing quality infrastructures match poorly to distributed modes of production (for example, FDA approval must be re-done by every manufacturer that wants to download and make a copy of even simple products for their local market - even if they only make it once). OpenDQM is guided by the principle that quality management in distributed manufacturing must be community-defined and context-sensitive. This position reflects the opinions of open hardware developers, manufacturers, clinicians, and users who operate across widely varying product classes, risk profiles, and regulatory environments. This is because uniform quality practices can exclude many manufacturers and creates an operating environment where only big producers can afford compliance or where regulatory bodies become overwhelmed. OpenDQM will therefore emphasize norms and best practices that emerge from practice, are informed by domain expertise, and are adaptable to specific contexts.

The ecosystem will prioritize transparency-based approaches to trust. Experience from open hardware communities and pandemic-era distributed manufacturing efforts demonstrate that transparency around design intent, provenance, testing, and responsibility can support rapid coordination, even in the absence of centralized oversight. OpenDQM will build on this experience by promoting approaches that make quality-relevant information auditable and interpretable across trust boundaries, including provenance, testing procedures, calibration data, failure modes, and manufacturing responsibility.

OpenDQM must complement existing regulatory and standards-based systems, rather than replace them. The ecosystem will engage with regulators, certification bodies, and standards organizations to understand how distributed quality practices can operate upstream of, alongside, or in preparation for, formal regulatory pathways. Thus, while still aspiring to meet safety and performance expectations, this approach will reflect the practical reality that many distributed manufacturing efforts currently operate outside traditional certification regimes.

Interoperability and reuse are central to the OpenDQM approach. OpenDQM emphasizes open standards, shared vocabularies, and modular approaches that allow distributed manufacturers to use these quality assurance tools and practices without centralized control. The scoping will treat existing solutions, such as Global Distributed Tracking (GDT), as foundational components within this landscape, providing concrete points of reference for how technical systems can support distributed quality goals.

2.3 National and Societal Needs Addressed

Users increasingly rely upon distributed manufacturing to meet national needs that centralized manufacturing systems struggle to address, including rapid response to supply chain disruptions [8,9], localized production of specialized goods, and agile adaptation to changing technical requirements. However, without credible and scalable approaches to quality management, consumers cannot safely or economically realize these benefits. Thus, the scoping of OpenDQM as an OSE will address critical societal needs by enabling transparency-based quality assurance that supports consumer safety and public trust.

This is increasingly crucial as distributed manufacturing expands into safety-critical domains such as medical devices, diagnostics, assistive technologies, and emergency response equipment. By reducing reliance on centralized reputation and enabling the evaluation of quality evidence across trust boundaries, OpenDQM will lower barriers for small and agile firms to participate in regulated or safety-sensitive markets while preserving accountability. This shared-burden approach to quality management reduces duplicative effort, improves economic efficiency, and strengthens innovation capacity. At the same time, OpenDQM will enhance supply-chain resilience by supporting the rapid onboarding of new producers, the local adaptation of designs, and the verification of product quality without centralized control, capabilities shown to be essential during the COVID-19 pandemic and other disruptions. By enabling local adaptation without sacrificing transparency or safety, OpenDQM will support resilient, trustworthy manufacturing systems that can evolve in step with emerging technologies, materials, and production methods.

2.4 Anticipated Broader Impacts of the OSE

An open-source ecosystem for distributed quality management will enable the wider adoption of distributed and agile manufacturing by providing transparent, credible approaches to quality assurance suited to decentralized production. While large manufacturers rely on centralized, reputational quality systems, the scope of OpenDQM must address the needs of small, geographically distributed producers operating across a variety of products and risk profiles. By supporting shared norms and interoperable tools, the ecosystem will improve the safety, reliability, and trustworthiness of distributed manufactured products, enabling the creation of new markets where speed and adaptability drive a product's value, rather than scale alone. OpenDQM will reduce the burden of quality management, support local adaptation without loss of accountability, and strengthen supply-chain resilience through rapid onboarding new producers. The scope of the OSE will be the United States and align with U.S. regulatory frameworks, while remaining adaptable to other national jurisdictions.

2.5 Problem Being Addressed

Distributed quality management efforts are currently fragmented across small manufacturers, makers and makerspaces, open hardware communities, and emergency or humanitarian initiatives. While distributed manufacturing has a growing practitioner base, producers inconsistently apply important quality practices such as testing, calibration, and failure reporting on a project-by-project basis. The current culture can reward novelty rather than repeatability and quality, often because the barriers to effectively managing quality are high. Interest in DQM has increased in response to pandemic-era manufacturing, sudden supply chain disruptions, and concerns over product safety and counterfeiting. However, shared standards, vocabularies, and coordination mechanisms remain limited, leaving the ecosystem as yet unconsolidated.

This proposal addresses the need for a focused scoping effort, aimed at the formation of an OSE for quality management efforts across distributed manufacturing. The problem is that a lack of a shared forum and coordinating structure prevents practitioners from collectively addressing challenges such as defining quality expectations, managing variability introduced by distributed production, and evaluating safety and reliability across independently manufactured products. Existing tools and practices address individual aspects of quality, but they remain unaggregated and difficult to integrate, leaving quality management inconsistent and costly to scale. This work focuses by assembling a community-led process and defining distributed quality management as a shared problem. OpenDQM does not replace existing platforms or tools, but instead provides a locus for dialogue and coordination, enabling stakeholders to jointly identify challenges and practical approaches to trustworthy distributed manufacturing.

3. Project Team

PI Robert Read is the founder and President of Public Invention and the architect of Global Distributed Tracking (GDT), an open-source cryptographic software that multiple manufacturers currently use. He previously served as a Presidential Innovation Fellow and was a co-creator of 18F within the U.S. federal government, where he worked on digital infrastructure for public service. During the COVID-19 pandemic, he played a leading role in the global open-source response to emergency ventilator shortages and has conducted applied research on oxygen production systems in collaboration with NASA. Dr. Read has also co-hosted the 2025 Distributed Quality Management (DQM) webinar series, which convened practitioners and experts across manufacturing, healthcare, and policy domains [21]. His experience positions him to lead the technical and organizational scope of OpenDQM.

Co-PI Victoria Jaqua is the Product Owner of GDT and the Medical Community Lead of Open Source Medical Supplies (OSMS), an organization created during the COVID-19 pandemic to curate and disseminate high-quality documentation for the distributed production of PPE and medical supplies. OSMS functioned as a widely recognized reference point for quality documentation at a time when distributed manufacturing was rapidly scaling, an effort documented in the 2021 OSMS white paper Design | Make | Protect [7] which Ms. Jaqua co-authored. Ms. Jaqua co-hosted the 2025 DQM webinar series and brings deep experience in community coordination, documentation standards, and engagement with humanitarian and healthcare stakeholders. Her leadership ensures OpenDQM is grounded in real practices and user needs.

Public Invention will appoint a postdoctoral researcher with specialized experience in open-source hardware and standards development. They will contribute technical analysis and synthesis across quality management practices, evaluate quality requirements for safety-critical and regulated product domains, assess how existing tools and documentation practices support or hinder distributed quality assurance, and assist in community data collection and event-driven dissemination (including at events such as Fab26) as well as authoring and development of the distributed quality management typology. The postdoctoral researcher will also support coordination with practitioners and subject-matter experts to ensure that OpenDQM reflects real-world manufacturing constraints and regulatory considerations.

Public Invention will also integrate consultants with expertise in distributed manufacturing, regulatory systems, and law. They will support the development of a distributed quality management typology and assess alignment with existing standards and compliance pathways. Through OSMS and related networks, the PIs are well connected to health, humanitarian manufacturing, and regulatory communities, enabling the targeted selection of consultants with relevant expertise.

4. Proposed Activity

The project will scope an OSE called OpenDQM to advance the science of distributed systems, quality management and sociotechnical governance. It will develop the first ‘State of the Art’ insights into the landscape of existing approaches (like federated systems with cryptography and edge-computing, as demonstrated by GDT) and critical stakeholder perspectives into a report for dissemination.

In collaboration with the ecosystem participants, the scoping of OpenDQM will also develop a new typology for quality management. This research output will provide a common language for DQM and for the OSE, which could later result in new architectures for DQM. The project will consider and propose governance innovations that, say, enhance trust through transparency rather than traditional models of centralized regulatory compliance.

The project will also broadly advance research into distributed manufacturing and resilient supply chains, which the COVID-19 pandemic showed to be critical to our bodily and economic health as a society.

4.1 Ecosystem Discovery

The Phase I team will conduct structured ecosystem research to identify stakeholders and challenges in distributed quality management. It will further attempt to organize quality management efforts into a more cohesive community of practice. This process will include key informant interviews and snowball sampling to find relevant efforts across open hardware communities, nonprofits, manufacturers, regulators, and tool developers. In parallel, the project will conduct literature and grey-literature reviews, including academic publications, blogs, GitHub repositories, and community documentation, to assess the current state of practice. Insights from these activities will inform the typology development of distributed quality management approaches, which community input will validate and stress-test across a variety of manufacturing contexts.

OpenDQM will promote community engagement through a sequence of online events leading to the selection of approximately ten participants for an in-person workshop at Fab26, complemented by a week-long challenge and hands-on workshops at Fab26 and Maker Faire. The discovery phase will include a dedicated workshop during the planned 2026 Public Invention Conference (PubInvCon) in Austin, TX. OpenDQM will demonstrate Global Distributed Tracking (GDT) to elicit feedback from makers and consumers, and to broaden conversations about quality priorities and trade-offs in distributed manufacturing. A communications role will prepare materials and coordinate participation and publication of a state-of-the-art synthesis of community findings and the proposed typology released as free online resources hosted on the OpenDQM project site.

4.2 Organization and Governance

A key element of the proposed work is to explore robust governance and organizational structures to support the scope of an OpenDQM OSE. Public Invention holds the credibility in relevant communities to perform this activity, as a 501(c)(3) non-profit with a track record of scholarship, of leadership in the open-source community, and of support to DQM initiatives (such as by being the fiscal sponsor of the GDT software). Interviews with, and research into, the governance and organizational structures of standards bodies, regulatory bodies, managing organizations for other open-source products, other systems using distributed quality management (such as in the food industry) or even community governance such as Decentralized Autonomous Organizations will inform the scope OpenDQM's potential structures. It is likely that the results may give scope for a single organization, a family of organizations or for frameworks for, say, dispute resolution, standards setting or resource allocation to administer an OpenDQM ecosystem.

4.3 Risk Analysis / Security

The project faces manageable risks related to coordination and scope. Tight timelines and reliance on external events may introduce scheduling challenges, and identifying suitably qualified consultants within the project period may require flexibility. There is also a risk that available data may be insufficient to support a fully developed typology of distributed quality management practices. The project will mitigate these risks by drawing on parallel domains, such as data quality management, and by engaging experienced practitioners within quality assurance, regulatory compliance, and ethics.

Effective stakeholder engagement is essential to identifying security, safety, and governance concerns. OpenDQM will integrate relevant expertise to address privacy, regulatory compatibility, and cross-jurisdictional considerations, including international requirements such as General Data Protection Regulation (GDPR). The ecosystem will also consider risks associated with collective action, such as cases where the failures of individual producers could affect broader adoption and trust.

Technical risks are limited, as Phase I does not commit to deploying new infrastructure. Several existing tools, such as Global Distributed Tracking (GDT), already illustrate the benefits of cryptography and addresses common web-based vulnerabilities. But risk mitigations will be in the OpenDQM scope.

4.4 Community Building

Engagement and onboarding of community stakeholders will begin by following up on Public Invention's earlier webinars on DQM. A series of online events will lead to a in-person workshops, as described above. Communications will be active throughout the project, including through a dedicated project website (with a domain name and hosting provided by a project collaborator, unfunded). The validation of the typology will increase engagement, as will inclusion of existing projects and approaches in the 'State of the Art' report. The launch of that report will further bring opportunities to build the community – and connect more stakeholders with each other.

4.5 Evaluation

Phase I evaluation will focus on ecosystem research and early convergence. Key metrics to track include participation in community engagement activities, evidence collection of emerging shared practices, and a curated list of between forty to one hundred early actors whose practices will anchor the distributed quality management typology. Existence proofs and review feedback / validation of the State-of-the-Art report and typology will be recorded. The PI may seek unfunded third-party reviews of project outputs to improve quality and readability etc. These metrics will also inform priorities and help assess OSE readiness.

5. Intellectual Merit

DQM is a rigorous technical challenge due to its intersection with novel, rapidly emerging economies, the challenge of evaluating hardware products that may interact with vulnerable populations (such as children or impaired populations), and the expanding complexity of hardware designs and manufacturing capabilities under development. With challenges widely recognized in industries like counterfeit luxury goods expanding into medical goods and pharmaceuticals [15]-[19]. The project advances the science of distributed systems, quality management and sociotechnical governance. It will develop the first 'State of the Art' insights, a new typology, new architectures for federated quality assurance, governance innovation for OSEs and will advance research into distributed manufacturing and resilient supply chains, particularly research into widespread PPE production in the COVID-19 response.

6. Broader Impacts

An OpenDQM OSE will improve the safety, trust, viability and reach of distributed manufacturing. It will positively impact: the Open Hardware movement; local and distributed manufacturers, and all the users and consumers who benefit from them, and; the quality management sector – particularly government regulators who struggle to stay ahead. It may also impact traditional manufacturers with new, open innovations in quality systems. Overall, massive numbers of small manufacturers could be positively impacted if the challenge of distributed quality management can be met, which would support new high-value jobs. And of course, it will ensure consumer, public and environmental safety in the new paradigm of production. The proposed work will strengthen the resilience of supply chains [20] which aligns distributed manufacturing communities with U.S. economic and national resilience goals.

7. Results from Prior NSF Support [No prior NSF support]

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SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION PUBLIC INVENTION				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Robert Read				PROPOSAL NO. 2617665	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Robert Read - Principal Inv				0.75			11,250
2. Victoria Jaqua				2.0			24,000
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (2) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				2.75			35,250
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (1) POST DOCTORAL SCHOLARS				4.0			48,000
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							15,000
4. (3) UNDERGRADUATE STUDENTS							43,200
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							141,450
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							0
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							141,450
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$10,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							44,700
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 5,000							
2. TRAVEL 28,500							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (30) TOTAL PARTICIPANT COSTS							33,500
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							1,000
3. CONSULTANT SERVICES							37,500
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							38,500
H. TOTAL DIRECT COSTS (A THROUGH G)							258,150
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Indirect Costs (Rate: 15.0, Base:258150)							
TOTAL INDIRECT COSTS (F&A)							38,723
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							296,873
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							296,873
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PD NAME Robert Read				FOR NSF USE ONLY			
ORG. REP. NAME* Robert L Read				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION PUBLIC INVENTION				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Robert Read				PROPOSAL NO. 2617665	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Robert Read - Principal Inv				0.75			11,250
2. Victoria Jaqua				2.0			24,000
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)							
7. (2) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				2.75			35,250
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (1) POST DOCTORAL SCHOLARS				4.0			48,000
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							15,000
4. (3) UNDERGRADUATE STUDENTS							43,200
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							141,450
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							0
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							141,450
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$10,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							44,700
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 5,000							
2. TRAVEL 28,500							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (30) TOTAL PARTICIPANT COSTS							33,500
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							1,000
3. CONSULTANT SERVICES							37,500
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							38,500
H. TOTAL DIRECT COSTS (A THROUGH G)							258,150
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							38,723
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							296,873
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							296,873
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PD NAME Robert Read				FOR NSF USE ONLY			
ORG. REP. NAME* Robert L Read				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

Budget Justification

Public Inventions' fiscal year is 1/1 to 12/31. Project duration is 12 months in the 2026 and 2027 fiscal years. Public Invention is a 501c3 non-profit organization.

Section A, Senior/Key Personnel (\$35,250)

Dr. Robert Read, PI

Funds of \$11,250 are requested to support the PI for 15 working days at \$750 per day with no fringe benefits. Role is high-level strategic leadership and ensuring the high standards of the work, in particular: engaging with standards bodies and certification organizations; reviewing and validating key outputs such as the State of the Art report, governance models and the final Scoping report; representing the project to NSF, and; POSE training and Fab26 participation. Rate is less than the likely 75th percentile salary for SOC Code 11-1011 for Chief Executives for the Austin-Round Rock-San Marcos, TX, area taken from <https://data.bls.gov/oes/#/area/0012420>, where the mean is \$287,780 so is \$1,107 per day in a 260 working day year (note that the 75% percentile salary figure for this SOC Code is not given).

Victoria Jaqua, Co-PI

A.2 Funds of \$24,000 are requested to support the Co-PI for 40 working days at \$600 per day with no fringe benefits. Role will focus on market assessment and ecosystem stakeholder discovery, including: being an ambassador and networker for the project; taking part in interviews; content search for the State of the Art report; hosting online and in-person events; advising on governance models for the OSE; reviewing the final Scoping work, and; POSE training participation, Maker Faire workshop delivery and Fab26 participation. Rate is less than the 75th percentile salary for SOC Code 11-2021 Marketing Managers for the Austin-Round Rock-San Marcos, TX, area from <https://data.bls.gov/oes/#/area/0012420>, which is \$204,990 so is \$788 per day in a 260 working day year.

Section B, Other Personnel (\$106,200)

Avinash Baskaran, PostDoc

Funds of \$48,000 are requested to support the PostDoc for 80 working days at \$600 per day with no fringe benefits. Role is responsible for project operations, co-ordination, authoring and stakeholder engagement, including: project systems; recruiting and co-ordinating project consultant and students; lead author of the State of the Art report; supporting data collection and synthesis for the feasibility and governance explorations; evaluating progress; contributing to and refining the final Scoping, and; POSE training participation and Fab26 participation. Rate is less than the 75th percentile salary for SOC Code 11-9041 Architectural & Engineering Manager for the Austin-Round Rock-San Marcos, TX, area taken from <https://data.bls.gov/oes/#/area/0012420>, which is \$214,860 so is \$826 per day in a 260 working day year. The PostDoc line in the budget is being used due to the qualifications of the chosen team member.

Graduate Student to be recruited

Funds of \$15,000 are requested to support an events and communications officer that may have completed a degree course for 50 working days at \$300 per day. Role is promoting awareness and supporting the documentation of the project, planning and developing materials for the workshops and engagements at Fab26, contributing to the design and preparation of the State of the Art report, and includes POSE training participation. Rate is less than the 75th percentile salary for SOC Code 27-3031 Public Relations Specialists for the Austin-Round Rock-San Marcos, TX, area taken from <https://data.bls.gov/oes/#/area/0012420>, which is \$80,170 so is \$308 per day in a 260 working day year.

2 x Undergraduate Students to be recruited

Funds of \$43,200 are requested to support three paid summer student interns who may be studying for a degree at \$14,400 each for up to 12 weeks (60 days) at \$1,200 per week (\$240 per day). Roles will include: interview support; quality management innovation discovery and documentation; literature reviews; learning from and documenting comparisons with other highly distributed quality management systems such as in the food industry, and; other work as may be assigned. Rate is less than the 75th percentile salary for SOC Code 13-1161 Market Research Analysts and Marketing Specialists for the Austin-Round Rock-San Marcos, TX, area taken from <https://data.bls.gov/oes/#/area/0012420>, which is \$103,010 so is \$396 per day in a 260 working day year.

Section D.1, Domestic Travel (\$39,000)PI, Co-PI & PostDoc Travel

Funds of \$6,000 are requested for three domestic trips of up to \$2,000 each to support essential project activities. Travel will be used for in-person meetings, stakeholder engagement (such as interviews or other engagements with regulators or standards bodies) or, if needed, coordination with partners, collaborators or consultants. Because specific meeting locations and dates are not yet known, these funds are budgeted as flexible domestic travel to enable timely participation in events that are critical to ecosystem development, community engagement and the Scoping output. All travel will comply with institutional policies and NSF requirements, and costs will be limited to economy airfare, standard lodging, per diem and ground transportation.

Maker Faire Travel

Funds of up to \$15,000 are requested for five project team members to travel to the Maker Faire Bay Area event to deliver workshops based on the Globally Distributed Tracking open-source product software and to collect maker and consumer feedback on distributed quality management.

<u>Item</u>	<u>Basis</u>	<u>Amount</u>	<u>Qty</u>	<u>Total</u>
Hotel for 4 nights	Prevailing market rates in Bay Area near Mare Island venue for Maker Faire workshop venue, using advance booking and no breakfast. 4 nights for workshop engagement activities throughout the 3-day Maker Faire event. Found using online hotel only comparison sites set to the dates of the Maker Faire conference.	\$300	5 x 6	\$9,000
Domestic flights	Prevailing market rates based on return economy airfares between Austin, TX and San Francisco, CA.	\$600	5	\$3,000
Per Diem	Breakfast, lunch and dinner, local travel based on estimates made online of local prevailing market rates in cafes etc (likely to be food trucks).	\$100	5 x 6	\$3,000

Fab26 Travel

Funds of \$18,000 are requested to support the PI, Co-PI, PostDoc and Graduate Student to attend the Fab26 conference at MIT in Cambridge, MA to engage with project stakeholders, conduct in-person interviews, run a workshop activity to collect data throughout the Fab26 event and hold dedicated project-specific workshops as part of the conference programme.

<u>Item</u>	<u>Basis</u>	<u>Amount</u>	<u>Qty</u>	<u>Total</u>
Fab26 Registration	Based on previous annual FabLab event fees (Fab25 charged about \$560 per ticket).	\$600	4	\$2,400
Hotel for 6 nights	Prevailing market rates in Cambridge, MA nearby MIT campus for Fab26 venue and workshop venue, using advance booking and no breakfast. 6 nights for workshop engagement activities throughout the full Fab26 event. Found using online comparison sites set to the dates of the Fab26 conference.	\$450	4 x 6	\$10,800
Domestic flights	Prevailing market rates based on return economy airfares between Austin, TX and Boston, MA.	\$600	4	\$2,400
Per Diem	Breakfast, lunch and dinner, local travel based on estimates made online of local prevailing market rates in student cafes etc.	\$100	4 x 6	\$2,400

Section D.2, Foreign Travel (\$5,700)

Funds of \$5,700 are requested to support an international guest to attend the Fab26 conference at MIT in Cambridge, MA to bring a global perspective to project activities, and to engage with other project stakeholders, contribute and discuss data and research, be interviewed and join the dedicated project-specific workshops as part of the conference programme. Public Invention is unlikely to be able visa applications for travel to the US, and therefore the costs of visas, but if these become a necessity then we will draw on our indirect costs to cover these.

<u>Item</u>	<u>Basis</u>	<u>Amount</u>	<u>Qty</u>	<u>Total</u>
Fab26 Registration	Based on previous annual FabLab event fees (Fab25 charged about \$560 per ticket).	\$600	1	\$1,800
Hotel for 6 nights	Prevailing market rates in Cambridge, MA nearby MIT campus for Fab26 venue and workshop venue, using advance booking and no breakfast. 6 nights for workshop engagement activities throughout the full Fab26 event. Found using online comparison sites set to the dates of the Fab26 conference.	\$450	6	\$2,700
Domestic flights	Prevailing market rates based on an approximate average of various return economy intercontinental airfares to Boston, MA, around the dates of the Fab26 event (noted as peak summer travel, World Cup travel season etc). Found using online travel comparison websites.	\$1,800	1	\$1,800
Per Diem	Breakfast, lunch and dinner, local travel based on estimates made online of local prevailing market rates in student cafes etc.	\$100	1	\$600

Section E.2, Participant/Trainee Support Costs - Stipend (\$5,000)

Funds are requested for stipends of up to \$250 each for up to 20 participants attending online interviews and events during the project. The stipends will support participant preparation, participation and follow-up activities associated with interviews, online workshops and community engagement events. These individuals will participate as learners and contributors and will not provide deliverables or services to the project team.

Section E.3, Participant/Trainee Support Costs - Travel (\$28,500)

Funds of \$28,500 are requested to support 10 participants to attend the Fab26 conference at MIT in Cambridge, MA to engage with project stakeholders, conduct in-person interviews, run a workshop activity to collect data throughout the Fab26 event and hold dedicated project-specific workshops as part of the conference programme. Participants will be asked to join for 2 full days (but are likely to be required to pay for full price Fab26 tickets).

<u>Item</u>	<u>Basis</u>	<u>Amount</u>	<u>Qty</u>	<u>Total</u>
Fab26 Registration	Ticket price based on previous annual FabLab event fees (Fab25 charged about \$560 per ticket).	\$600	10	\$6,000
Hotel for 3 nights	Prevailing market rates in Cambridge, MA nearby MIT campus for Fab26 venue and workshop venue, using advance booking and no breakfast. 6 nights for workshop engagement activities throughout the full Fab26 event. Found using online comparison sites set to the dates of the Fab26 conference.	\$450	3 x 10	\$13,500
Domestic flights	Prevailing market rates based on return economy airfares between Austin, TX and Boston, MA.	\$600	10	\$6,000
Per Diem	Breakfast, lunch and dinner, local travel based on estimates made online of local prevailing market rates in student cafes etc.	\$100	3 x 10	\$3,000

Section F.1, Publication Costs (\$1,000)

Funds of \$1,000 are requested to cover: printing costs for project documents; workshop materials including for Fab26 and the Maker Faire; printing copies of the State of the Art report, and; very minor publication related costs such as subscriptions, ISBN purchases, online services or fees.

Section F. 3 Consultant Services (\$37,500)

Consultants will be procured / recruited during the project, so the individuals are to be determined.

Final times and rates will be determined during the project and be based on project need at the time.

The basis of the budgeted rates is Public Invention's previous experience and actual rates will be subject to negotiation and request for 'non-profit rates' from the consultants. The amounts below are maximum expected rates – if the rates are lower then the project may request more time, but the rates will not be permitted to go higher. In Public Invention's experience – and with legal consultants in particular – consultants may bill by the hour rather than the day so partial days are to be expected.

No consultant will serve in a role similar to senior personnel. No single consultant will receive more than \$50,000 over the entire award period.

Funds are requested as follows:

<i>Individual Consultant</i>	<i>Role, to provide expertise on..</i>	<i>Max Day</i>	<i>Max Rate</i>	<i>Total</i>
Innovation & Strategy	Independent analysis of results; workshops leader & contributor;	40	\$750	\$30,000
Quality Systems Expert	FDA medical device regulations; wider / international regulations.	5	\$750	\$3,750
Liability Legal Advisor	Liability; Consumer protection; Safety; Warranty Agreements; etc.	5	\$750	\$3,750

None

IDENTIFYING INFORMATION:

NAME: Read, Robert

ORCID iD: <https://orcid.org/0000-0003-4749-7045>

POSITION TITLE: President and Founder

PRIMARY ORGANIZATION AND LOCATION: Public Invention, a US 501c3 Public Charity, Austin, Texas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Texas (Austin), Austin, Texas, United States	PHD	08/1988 - 12/1995	Computer Science
Rice University, Houston, Texas, United States	BA	08/1983 - 12/1988	Computer Science

Appointments and Positions

2015 - present President and Founder, Public Invention, a US 501c3 Public Charity, Austin, Texas, United States

2013 - 2015 Presidential Innovation Fellow, Director of 18F Consulting, General Services Administration, Washington, District of Columbia, United States

2008 - 2013 Director of Product Development, Planiview, Inc., Austin, Texas, United States

Products

Products Most Closely Related to the Proposed Project

1. Alice Villar, Sarah Abowitz, Robert Read, James Butler. Maximizing Supply Chain Resilience: Viability of a Distributed Manufacturing Network Platform Using the Open Knowledge Resilience Framework. Operations Research Forum. 2024 March. DOI: 10.1007/s43069-024-00303-1
2. Robert L. Read, Nathaniel Bechard, Victor Suturin, Antal Zuiderwijk, Michelle Mellenthin. The PolyVent educational platform: An open mechanical ventilation platform for research and education. HardwareX. 2025 March. DOI: 10.1016/j.ohx.2024.e00615
3. Robert Read, Victoria Jaqua, Christina Cole, Francesca Williams, Nina Lahoti, Nora Moor. Global Distributed Tracking of Supplies via Free-libre Open Source Cryptography Obviating Authentication. 2025 June. DOI: 10.31224/4732
4. Robert L. Read, Christina A. Cole, Victoria Jaqua, Andrew Lamb. Open Know-How and an Open Source Hardware Registry as a Unifying Discoverability Mechanism for Open Source Hardware Humanitarian Engineering Projects. ; c2024. DOI: 10.1109/GHTC62424.2024.10771512
5. Robert L. Read, Victoria Jaqua, Christina A. Cole, Francesca Williams, Nina Lahoti, Nora Moor. Global Distributed Tracking of Supplies via Free-Libre Open Source Cryptography Obviating Authentication. ; c2025. DOI: 10.1109/GHTC66843.2025.11266358

Other Significant Products, Whether or Not Related to the Proposed Project

1. Read R. Calculating the Segmented Helix Formed by Repetitions of Identical Subunits thereby Generating a Zoo of Platonic Helices. Mathematics. 2022 July 21; 10(14):2533-. Available from: <https://www.mdpi.com/2227-7390/10/14/2533> DOI: 10.3390/math10142533
2. L. Read Robert, Baskaran Avinash. Controlling a variable geometry truss tetrobot with an isomorphic puppet controller. 2020 2nd International Conference on Artificial Intelligence, Robotics and Control; 2020; c2020.
3. Read R. Transforming Optimal Tetrahelices Between the Boerdijk–Coxeter Helix and a Planar-Faced Tetrahelix. Journal of Mechanisms and Robotics. 2018 October 01; 10(5):- . Available from: <https://asmedigitalcollection.asme.org/mechanismsrobotics/article/doi/10.1115/1.4040433/474081/Transforming-Optimal-Tetrahelices-Between-the> DOI: 10.1115/1.4040433
4. Half-round total internal reflection magnifying prism. US. 2005 November.
5. Mapping defects or dirt dynamically affecting an image acquisition device. US. 2000 March.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Read, Robert in SciENcv on 2026-01-07 19:16:31

Effective 05/20/2024

NSF BIOGRAPHICAL SKETCH

OMB-3145-0279

IDENTIFYING INFORMATION:

NAME: Jaqua, Victoria Finch

ORCID iD: <https://orcid.org/0009-0006-9907-5150>

POSITION TITLE: Medical Community Lead

PRIMARY ORGANIZATION AND LOCATION: Open Source Medical Supplies, Washington DC, District of Columbia, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Kennebec Valley Community College, Fairfield, Maine, United States	AA	01/2005 - 05/2008	Radiologic Technology

Appointments and Positions

- 2020 - present Medical Community Lead, Open Source Medical Supplies, Washington DC, District of Columbia, United States
- 2022 - present Steering Committee Member, Global Open Source Quality Assurance System, Austin, Texas, United States
- 2022 - present Growth Partner, Glia, London, Ontario, Canada
- 2021 - present Board Member, Public Invention, Austin, Texas, United States

Products**Products Most Closely Related to the Proposed Project**

1. Robert L. Read, Victoria Jaqua, Christina A. Cole, Francesca Williams, Nina Lahoti, Nora Moor. Global Distributed Tracking of Supplies via Free-Libre Open Source Cryptography Obviating Authentication. ; c2025. DOI: 10.1109/GHTC66843.2025.11266358
2. Robert L. Read, Christina A. Cole, Victoria Jaqua, Andrew Lamb. Open Know-How and an Open Source Hardware Registry as a Unifying Discoverability Mechanism for Open Source Hardware Humanitarian Engineering Projects. ; c2024. DOI: 10.1109/GHTC62424.2024.10771512
3. Dawei Liu, Apoorv Kulkarni, Victoria F. Jaqua, Christina A. Cole, Joshua M. Pearce. Distributed manufacturing of an open-source tourniquet testing system. HardwareX. 2023 September. DOI: 10.1016/j.ohx.2023.e00442

Other Significant Products. Whether or Not Related to the Proposed Project

1. Jaqua V, Cole C, Cavalcanti G, Merlo S, Deml T, Emery K, Rubenstein MW. Design | Make | Protect. Open Source Medical Supplies website. 2021 January. Available from: https://opensourcemedicalsupplies.org/wp-content/uploads/2021/01/Design-Make-Protect_21.01.27.pdf
2. Jaqua V. Why a Global Open Source Quality Assurance System (GOSQAS)? Motivating Solutions for Diverse Use-Cases. GOSQAS GitHub. 2023. Available from: <https://github.com/gosqasorg/home/blob/main/GOSQAS%20Use%20Case.pdf>

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Jaqua, Victoria Finch in SciENCv on 2026-01-07 19:56:11

Other Personnel Biographical Information

Data Not Available

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Read, Robert Lee

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: <https://orcid.org/0000-0003-4749-7045>

*POSITION TITLE: President

*ORGANIZATION AND LOCATION: Public Invention, Austin, Virtual, USA

Proposals/Active Projects

*Proposal/Active Project Title: COG System Control Evaluation

*Status of Support: Current

Proposal/Award Number: JETSII SOW: PRS26-25

*Source of Support: NASA

*Primary Place of Performance: Public Invention

*Proposal/Active Project Start Date: (MM/YYYY): 04/2026

*Proposal/Active Project End Date: (MM/YYYY): 10/2026

*Total Anticipated Proposal/Project Amount: \$44,965

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2026	3

*Overall Objectives:

This effort is to explore the possibility of a high-frequency (HF) heating system for a ceramic oxygen generator.

*Statement of Potential Overlap:

There is no overlap.

*Proposal/Active Project Title: NASA Oxygen Concentrator (subcontract) JETSII SOW: PRS25-5179

*Status of Support: Current

Proposal/Award Number: JETSII SOW: PRS25-5179 (Subcontract)

***Source of Support:** NASA

***Primary Place of Performance:** Public Invention

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** \$43,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	4

***Overall Objectives:**

This is a subcontract given to Public Invention to research new ideas for a solid-state (no moving parts) ceramic oxygen concentrator.

***Statement of Potential Overlap:**

There will be no overlap in time or conflict of interest with this project.

***Proposal/Active Project Title:** POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** Public Invention

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2026

***Proposal/Active Project End Date: (MM/YYYY):** 07/2027

***Total Anticipated Proposal/Project Amount:** \$258,150

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2
2027	2

***Overall Objectives:**

This proposal.

***Statement of Potential Overlap:**

This proposal.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Read, Robert in SciENCv on 2026-05-06 10:57:58

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Jaqua, Victoria

*POSITION TITLE: Medical Community Lead

*ORGANIZATION AND LOCATION: Open Source Medical Supplies, Washington, District of Columbia, United States

Proposals/Active Projects

*Proposal/Active Project Title:

POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management

*Status of Support:

Pending

Proposal/Award Number:

*Source of Support:

NSF

*Primary Place of Performance:

Public Invention

*Proposal/Active Project Start Date: (MM/YYYY):

08/2026

*Proposal/Active Project End Date: (MM/YYYY):

07/2027

*Total Anticipated Proposal/Project Amount:

\$258,150

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2026	1
2027	0.3

*Overall Objectives:

This proposal

*Statement of Potential Overlap:

This proposal

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of

the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Jaqua, Victoria in SciENcv on 2026-02-08 13:28:26

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Read, Robert L.	Public Invention	01/01/26

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R				

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Donald S. Fussell	The University of Texas at Austin	
T	Abraham (Avi) Silberschatz (no middle initial.)	The University of Texas at Austin	

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
C	Cole, Christina	Open Source Medical Supplies	christina@opensourcemedicalsupplies.org	01/01/26
A	Baskaran, Avinash	Auburn University	azb0180@auburn.edu, mechanical engineering	01/01/26
C	Lamb, Andrew	Internet of Production Alliance	andrew.lamb@internetofproduction.org	01/01/26
A	Bechard, Nathaniel	Public Invention		01/01/26
A	Suturin, Victor	Public Invention		01/01/26
A	Zuiderwijk, Antal	Public Invention		01/01/26
A	Mellenthin, Michelle	Colorado Mesa University	mmellenthin@coloradomesa.edu	01/01/26
A	Pearce, Joshua	Western University	joshua.pearce@uwo.ca	01/01/26
A	Kincheloe, Lawrence	Public Invention		01/01/26
A	Lee, Forrest E.	Public Invention		01/01/26

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B				
E				

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Jaqua, Victoria F.	Open Source Medical Supplies	01/01/26

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Cole, Christina	Open Source Medical Supplies	christina@opensourcemedicalsupplies.org	01/01/26
A	Read, Robert	Public Invention	read.robert@pubinv.org	01/01/26
C	Baskaran, Avinash	Auburn University	azb0180@auburn.edu, mechanical engineering	01/01/26
A	Lamb, Andrew	Internet of Production Alliance	andrew.lamb@internetofproduction.org	01/01/26
A	Lahoti, Nina	Global Open Source Quality Assurance System	hello@ninalahoti.com	01/01/26
A	Moor, Nora	Global Open Source Quality Assurance System	noramoor2025@gmail.com	01/01/26
A	Williams, Francesca	Global Open Source Quality Assurance System	francescagayle97@gmail.com	01/01/26
A	Pearce, Joshua	Western University	joshua.pearce@uwo.ca	01/01/26
A	Kulkarni, Apoorv	Western University	apoorv.kulkarni@uwo.ca	01/01/26
A	Liu, Dawei	Western University	dliu428@uwo.ca	01/01/26

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B				
E				

Synergistic Activities – Co-PI Victoria Jaqua

1) **Medical Community Lead:** In my role as Medical Community Lead for Open Source Medical Supplies (OSMS), I interface with diverse medical professionals through various relationship stages to open source medical hardware. During 2020, I matched medical facilities to donated PPE supplies made from open source designs, helped medical professionals develop their open source designs through community interface, and documented medical innovation in the OSMS/Nation of Makers (NOM) white paper, Design | Make | Protect. I co-host and panel medical and humanitarian community events, including RespiraCon II (2022), National Academies Personal Protective Equipment and Personal Protective Technology Product Standardization for a Resilient Public Health Supply Chain: A Workshop (2023), Distributed Quality Management: Show & Tell (2025), SolarPunk HealthCare Conference (2025), Global Health Workshop 2025, and am a regular presenter at Public Invention's monthly Inventor's Gatherings. I also facilitate medical and humanitarian response to disaster and conflict zones through numerous partners in the NGO sector.

2) **Open Source Design Curation:** Since 2020, I have been the lead curator for the Open Source Medical Supplies Project Library. I manually source open source designs through community discovery, relationship building, and project management. I review open source designs for licensing, documentation completeness, development rigor, and research integration. My curation experience has allowed me to consult on broader open source hardware discoverability projects, such as Open-Know-How and related search engine construction, and participate in the Open Source Hardware Association (OSHWA) Healthware 2025 workshops.

3) **Product Owner:** As a Steering Committee member for the Global Open Source Quality Assurance System (GOSQAS) since 2022, I am the Product Owner for Global Distributed Tracking (GDT), the flagship GOSQAS open source software that enables close-looped tracking for products, information, and logistics. I represent the GDT user and guide the development team on issue and launch prioritization. I also oversee GDT team onboarding, product marketing, customer support, impact communications, market assessment and growth strategy. GDT users I have onboarded and managed include Glia, Distributed Medical Device Manufacturing (DMDM), and Médecins Sans Frontières (MSF).

4) **Project Manager:** I am the Project Manager for the open source Glia tourniquet, and facilitate diverse open source hardware development through varying stages. I manage peer-reviewed publication development (two papers currently in progress) and connect development stakeholders to burgeoning open source projects. Organizational partnerships as a PM include ASTM, Western University, Imperial College London, Air Support Project, Public Health Action Network, and MAKE Magazine.

5) **Growth & Strategic Partnerships Lead:** I serve as a growth partner for the open source and humanitarian community as a whole by networking, introducing, establishing partnerships, sharing resources, and capacity building via fundraising and marketing strategy.

Synergistic Activities – PI Robert L. Read

Leadership of 10 projects at Public Invention

Robert L. Read created Public Invention, a US 501c3, in 2018, and has led it since then. Public Invention is one of the few independent research non-profit organizations that produces significant humanitarian engineering research. In addition to executive leadership, Rob acts as the Head Invention Coach to 10 open-source projects. These have resulted in about a dozen peer-reviewed papers in the last 8 years, representing the first publication for many of our volunteer mentees, boosting their research careers. Rob personally mentors about 20 volunteers and lately has overseen and trained new Invention Coaches who also mentor volunteers.

All of these projects are open source. Significant accomplishments include the creation of an open-source pandemic ventilator (the PolyVent), a ventilator spirometer (the VentMon), the completely novel passive ferrofluid check-valve, and Global Distributed Tracking. Public Invention is pioneering open-source humanitarian invention outside of both for-profit companies and Universities.

Architect of Global Distributed Tracking (GDT)

Rob is the chief architect and inventor of Global Distributed Tracking, a free-libre open-source software system. This software can be used by anyone to host a website that provides encrypted tracking in a new model which implements a security model which does not require authentication-a major advantage for international operation in LMICs.

GDT is Public Invention's most important completely free "product". It is currently be used to globally track medical devices by two external non-profits, Glia and Distributed Medical Device Manufacturing, and we are working with Médecins Sans Frontières on a new project.

Oxygen Concentration for NASA

Public Invention has designed a digital control system for a research oxygen concentrator created by NASA and American Oxygen, which has been deployed by the Scottish charity Kids Operating Room as a demonstration global health technology.

Architect of Project Data

Rob is also a board member at Helpful Engineering, a US 501c3, where he is the architect of Project Data, an open-source software project to create resilient supply chains to mitigate the suffering caused by natural disasters. It represents an attempt to organize small firms to repair temporarily disrupted supply chains.

Data Management & Sharing Plan

In this Phase I project, a variety of datasets, documentation, visualizations and publications will be produced, archived and disseminated as part of the scoping and planning of the OSE.

None of the data used in the Global Distributed Tracking tool are sensitive. Since the purpose of the tool is to enable open sharing and transparency, user data is entirely consent-based. Data will be made available to the NSF with all reporting for this project, as per the NSF guidance on Technical Reporting Requirements. The data will be available to the public starting on May 1, 2026. Contact Robert L. Read at read.robert@pubinv.org (512-913-6103) for more information regarding this plan or data accessibility. Any and all future sub-awardees not identified in this plan will have as a condition of their contract acceptance of this data sharing plan.

Products of Research - *Types of Data Produced*

- *Technical Artifacts.* As a part of this work, we will develop various designs, specifications, images, documentation, publications, web content and educational content.
- *Interview and Survey Data.* The PI, Co-PI and PostDoc and graduate student, undergraduate students and consultants working on this project will conduct and analyze semi-structured interviews, surveys, and focus groups with likely OSE community members on discovery, governance and structures. All transcripts, recordings and analysis will be stored securely.

Security

Public Invention has good experience in managing data and secure, appropriate sharing. Recommendations provided by CISA's ESF for mitigation of threat to Identity and Access Management (IAM) are used as a general approach.

IAM Threat Mitigation Technique	Approach
Identity Governance	Our project will implement the JML (Join, Move, Leave) lifecycle process as a product/platform agnostic methodology to manage identity governance.
Environmental Hardening	Foundational information security measures are provided by the Public Invention team in combination with external cloud-based services, the maintenance of which includes resource auditing. For all data collected, documentation, and visuals generated by this project, the LOCKSS (lots of copies keeps stuff safe) or the 3-2-1 principles will be followed in the event of disk failure.
Identity Federation and SSO	Shibboleth SSO with MFA will be used to access all documentation related to this project.
Multi-Factor Authentication	MFA will be managed using independent authentication.
IAM Monitoring and Auditing	Best practices will be used as a framework for monitoring all project artefact production.

Metadata, Data Formats and Standards

To allow for the feasible dissemination and interoperability of our data and research outputs that are platform-agnostic, all data will be ASCII-encoded and we will use a variety of file formats, with preservation as both .txt and .pdf.

Written documentation such as governance bylaws, summary interview findings, catalogues of existing products, and open educational resources (OER) for instructional guidance will be created and preserved in .docx, .pdf, and .md/markdown.

Visual materials produced, such as process mapping using platforms such as Miro, LucidChart, or Figma, will be exported and preserved as .svg, .jpeg, and .pdf. Images that are part of our communications and outreach efforts will be produced and preserved as .jpeg and .png. All logos will be produced and preserved as .svg and .jpeg.

Use of metadata schemas for preservation and discoverability of project output documentation will be determined as outputs are produced, relying on recommendations provided by OSS and community-centered activities such as schema.org and W3C. Quality assurance process models will be done in accordance with ISO (9001 family, quality assurance) and IEEE.

Dissemination, Access, and Sharing of Data

The data and research outputs will be made available publicly and open source, using the Public Invention website, pages under the OpenDQM web domain, the Internet of Production Alliance website, open access journals, online community forums and potential OSE community member organization websites. Outputs of this project will be shared in publications such as the Journal of Open Hardware and ACM, and presented at conferences such as the IEEE Global Humanitarian Technology Conference (where the Global Distributed Tracking tool has previously been presented), FabX Global events (particularly Fab26 during the project timeline) and other appropriate open events.

The project will encourage the use of Public Invention Free-Culture License Guidelines, which specify different reciprocal or “share-alike” licenses for different kinds of content: software, hardware, documentation, scientific articles, freehand graphic art, diagrammatic graphic art and devices. Following generally accepted best practices, these use the “git” version control system, in free publicly readable repositories which allow the submission of improvements in the form of “pull requests” to be vetted by the project managers. The OpenSSF Security Best Practices will be encouraged by the managing organization of the ecosystem, and are universal in their applicability to code submissions.

Re-Use, Re-Distribution, and Derivatives

Access to research databases and other software tools will be available for educational, research, and non-profit purposes. This access will be provided using web-based user interfaces, but also the report on the ‘State of the Art’ using CC-BY-SA-4.0 for documentation or equivalents.

Archiving of Data

Co-PI will be responsible for all data management during and after collection and creation. All data generated over the course of this project will be preserved using the LOCKSS standard protocol of triple redundancy: (1) storage on local drive (2) storage on external portable drive (3) storage on cloud-based/online server. Cloud storage will be password protected/encrypted, and their selection requires them to have an automated, regular backups and archiving approach. All reports, presentations and research, and datasets with data presented in aggregate/deidentified from personal information identifiers, will be licensed CC-BY-SA-4.0.

Postdoctoral Researcher Mentoring Plan

One team member is currently working as a PostDoc researcher and, out of an abundance of caution, the project has included them under the PostDoc line in the budget. Public Invention is not a university but rather a 501c3 and so will not be offering a PostDoc position as such, but instead will offer an existing PostDoc a contract work in the project team. Nevertheless, Public Invention offers this mentoring plan.

The PostDoc's development will be enhanced through a program of mentoring activities. The goal of the mentoring program will be to provide the skills, knowledge and experience to prepare the PostDoc researcher to excel in their career path. Specific elements of the mentoring plan will include:

- Working with the PostDec researcher to establish and implement an Individual Development Plan.
- Seminars, workshops and individual consultations with the PI and Co-PI on data collection, preservation and dissemination.
- Opportunities to network with scholars and practitioners who are leaders in our field by networking with them when they participate in events.
- Travel to select conferences, e.g. Fab26 at MIT in Boston (travel funds are included in the budget), with the goal that the PostDoc will present at the conference.
- Participation in a monthly community calls held by Public Invention, in which speakers will be invited to discuss subjects related to their career development.

Access to office space and administrative services will be supported from Indirect Costs if needed.

The Principal Investigator is an experienced mentor and coach, and is the Head Coach of Public Invention's network of coaches.

We will expect the PostDoc to meet at least weekly with the PI and Co-PI. During these meetings, we will ask questions informed by our own experience of navigating research and community discovery projects.

Working on this research agenda will expose the PostDoc to the State-of-the-Art in several research areas/directions related to the proposed research. The PostDoc will be the lead author on the project's report on the State-of-the-Art in Distributed Quality Management. What they learn will broaden their education and give them additional tools to employ in their future career.

Robert Read Ph. D, Principal Investigator
Public Invention
1709 Norris Drive
Austin, Texas 78704

7th January 2025

Subject: Letter of Collaboration for Public Invention's NSF POSE Phase I proposal.

From: Distributed Medical Device Manufacturing collective, a GDT user and contributor.

Dear Dr. Read,

We intend to collaborate with you if the proposal submitted by Public Invention entitled "*POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management*" is selected for funding by NSF.

Distributed Medical Device Manufacturing (DMDM) is a collective focused on enabling the safe, effective, and responsible distributed production of medical devices. Our work requires navigating regulatory constraints, documentation requirements, provenance tracking, and quality assurance challenges in decentralized manufacturing contexts. As a result, we are frequently required to develop ad hoc solutions to address gaps in existing quality management systems that were not designed for distributed production. The establishment of an open, community-governed ecosystem for distributed quality management directly addresses barriers we encounter in practice and represents a critical step toward sustainable, scalable distributed medical manufacturing.

DMDM has been an active user and contributor to Global Distributed Tracking (GDT) by applying it to real-world distributed manufacturing workflows. Members of our community have also participated in and contributed to Distributed Quality Management discussions and events, including those convened through the Global Open Source Quality Assurance System (GOSQAS) and related gatherings such as the Solarpunk Healthcare Conference, helping surface practical requirements and constraints from the medical device domain.

If funded, DMDM intends to support the OpenDQM Phase I project by facilitating participation from our distributed network of medical device manufacturers and collaborators in OpenDQM interviews, workshops, and scoping activities, and sharing concrete lessons learned, unmet needs, and design constraints derived from operating in regulated and safety-critical manufacturing environments

Yours sincerely,



Michael Jacobs
Co-Founder, Operations
mike@dmdm.icu

Distributed Medical Device Manufacturing
www.dmdm.icu
PO Box 381, Tucson, AZ 85702



January 8, 2026

Robert Read Ph. D, Principal Investigator
Public Invention
1709 Norris Drive
Austin, Texas 78704

RE: Letter of Collaboration for Public Invention's NSF POSE Phase I proposal

Dear Dr. Read,

This letter is to confirm that we intend to collaborate with your project "*POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management*" if it is selected for funding by NSF. We want to ensure that our global network of 2,600 Fabrication Laboratories - known as FabLabs - would benefit from participation in the OpenDQM Ecosystem. Our work started with an NSF grant in 2001, which created the first FabLab here at MIT, so we are proof of the scale of adoption that can be achieved in this field.

The FabLabs network operates across education, research, entrepreneurship and community development. As FabLabs increasingly engage in the local production of functional, safety-critical, and research-relevant artifacts, challenges of ensuring quality and traceability across distributed manufacturing environments have become more prominent.

Members of the FabLab community have engaged with and promoted your Global Distributed Tracking (GDT) software as an emerging tool for provenance tracking and accountability within distributed manufacturing contexts, reflecting broader interest among FabLabs for quality management.

If funded, the Fab Foundation intends to provide opportunities for your project to engage with and gather ideas and feedback from FabLabs, mainly through our forthcoming global conference at MIT known as Fab26. These annual events bring together up to 3,000 FabLab practitioners, educators, researchers and policymakers from all over the world. Further, we will engage our communication channels and provide assistance in our event programming and conference venues to disseminate information about the OpenDQM project and to encourage participation in its scoping activities.

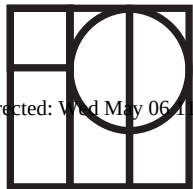
The Fab Foundation 50 Milk Street, 16th Floor Boston, MA 02109 USA +1 857-333-7777 1

Do not hesitate to contact us should you require any further information or assistance.

Yours sincerely,

A handwritten signature in blue ink, appearing to read "S. J. Lassiter", with a stylized flourish at the end.

Sherry J. Lassiter
President & CEO
Fab Foundation



INTERNET OF PRODUCTION

Corrected: Wed May 06 11:21:01 EDT 2026

Submitted/PI: Robert L. Read /Proposal No: 2617665

Robert Read Ph. D, Principal Investigator
Public Invention
1709 Norris Drive
Austin, Texas 78704

7th January 2025

Subject: Letter of Collaboration for Public Invention's NSF POSE Phase I proposal.
From: The Internet of Production, contributor to GOSQAS Global Distributed Tracking.

Dear Dr. Read,

If your proposal entitled "*POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management*" is approved for funding by the National Science Foundation, the Internet of Production Alliance intends to collaborate with you and your team to help achieve the objectives of your proposal.

The Internet of Production Alliance brings together people, initiatives and organizations around the world who are advancing open, distributed and locally resilient manufacturing systems. Our work focuses on open standards and infrastructures to enable large networks of local manufacturers to collaborate, which involves research, policy and community convening activities. Distributed quality management has been a central concern of our community, particularly as production systems move away just from centralized industrial models towards more globally networked local manufacturing.

Your project is a natural continuation and formalization of ideas that our alliance has helped surface through the recent webinars we have organized and co-hosted with you on Distributed Quality Management. These showcased the Global Distributed Tracking software alongside 25 other Alliance members working in this area.

Our collaboration can include gathering contributions from members of our alliance to your scoping activities, and we are happy to support discussions, workshops and outreach events at convenings like Fab26 (where we regularly participate). We will also provide dedicated pages to your project on our website and help plan how to fund and sustain the OpenDQM open-source ecosystem in the longer-term.

Overall, this is an important, urgent and exciting project and we and all the members of our alliance are ready to collaborate with you to support it in any way we can.

Please let me know if there is anything else we can do to support your proposal to NSF POSE.

Yours sincerely,



Andrew Lamb MEng MA Cantab MIET FRSA
Alliance Chair

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Robert Read Ph.D
Principal Investigator, Public Invention
1709 Norris Drive
Austin, Texas 78704

10th January 2026

Subject: Letter of Collaboration for Public Invention's NSF POSE Phase I proposal.

From: The Open Flexure Microscope & GitBuilding projects, potential users of GDT.

Dear Dr. Read,

I am writing to confirm our intention to collaborate with you on your proposed NSF project called *"POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management"* if it is selected.

The OpenFlexure Microscope project supports thousands of people in at least 67 countries who are making and using our designs and documentation to 3D print, assemble, and adapt microscopes for scientific research, educational use. A number of clinical partners and manufacturers are working with us to work towards certified production as a diagnostic microscope.

As you know, responsibility for the final quality of the microscopes ultimately lies with their makers, but we play an important role in making sure the documentation they get from us is excellent. This is why we created the open-source software called GitBuilding – we needed a way to do good documentation, and to ensure that all of our documentation for each variation of the microscope was always in sync, and fully up-to-date with the design. Since no good open-source solution existed for hardware documentation, we made our own. We see GitBuilding as playing a role in managing quality in distributed manufacturing contexts, and therefore are very keen to collaborate with you.

To continuously improve the microscope design we need quality control and calibration data from microscope builders around the world. At the same time, it is important that we do not embed telemetry directly into our documentation as this would infringe on the autonomy and rights of members of our communities. The challenge for us is determining how to obtain this data, from those in our community who wish to feedback the data in an open, secure, and fair way. Members of the OpenFlexure and GitBuilding communities have engaged with your Global Distributed Tracking (GDT) tool as one of several emerging approaches to address this challenge. Indeed, we were delighted to be able to showcase GitBuilding alongside GDT in your Distributed Quality Management webinars last year and look forward to further such events during the Scoping OpenDQM project.

Alongside documentation with GitBuilding, we have taken other measures to enhance community review and issue tracking. We will share our experiences related to documentation, provenance, and quality control practices in the distributed manufacture of the OpenFlexure Microscope and other open science hardware devices. Some of the other ways we will collaborate include participating as interviewees to

inform your stakeholder and needs assessment research, serve as early adopters and contributors to any emerging OpenDQM framework and participate in discussions and feedback sessions to test assumptions. We would also be glad to collaborate in the project's events and share its progress with our communities.

Yours sincerely

A handwritten signature in black ink, appearing to read "Julian Stirling".

Dr. Julian Stirling
Chief Executive

Robert Read Ph. D, Principal Investigator
Public Invention
1709 Norris Drive
Austin, Texas 78704

9th January 2025

Subject: Letter of Collaboration for Public Invention's NSF POSE Phase I proposal.
From: The Open Science Hardware Foundation, which is also a fiscal host for the Gathering for Open Science Hardware global community of scientific instrument makers.

Dear Dr. Read,

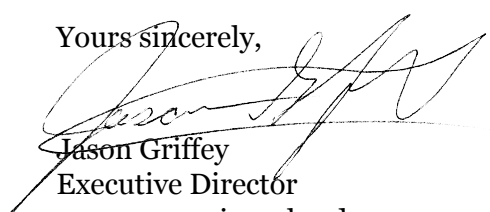
Should the National Science Foundation approve funding for the proposal submitted by Public Invention, titled "*POSE: Phase I: Scoping OpenDQM – an OSE for Distributed Quality Management*" we intend to collaborate with you to advance the project objectives.

The Open Science Hardware Foundation (OSHF) is a 501(c)3 organization that is dedicated to advancing open-source scientific hardware – making lab tools, instruments and research equipment freely accessible, modifiable and shareable worldwide. We act as both an institutional home and a capacity-builder for the Global Open Science Hardware (GOSH) movement, a global community of practitioners who design, build and deploy open scientific instruments across diverse institutional and geographic contexts. OSHF also maintains and promotes a definition of Open Science Hardware, including expectations around complete design files, documentation and licensing, which establishes a baseline for reproducibility and auditability. As Open Science Hardware continues to scale and is being adopted internationally, we seek to improve the reliability and quality of what gets made. This is why we want to collaborate with you.

Our community has recognized your Global Distributed Tracking software as a step toward effective distributed quality management. Through our workshop on "Understanding and Advancing Enabling Ecosystems for Open Science Hardware", we have sought to explore lines of inquiry that can catalyze trust and adoption of Open Science Hardware.

We will collaborate by circulating OpenDQM project activities to the very active GOSH community by including OpenDQM in the agenda of the meetings and events, and by assisting to facilitate data collection from GOSH community members in interviews and discussions forums. We will also share our experience as a host organization of an open-source ecosystem, and what is needed to scale and sustain.

Yours sincerely,



Jason Griffey
Executive Director

www.opensciencehardware.org

Open Science Hardware Foundation, Inc.

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info@opensciencehardware.org Page 47 of 50 www.opensciencehardware.org

List of Project Personnel, Collaborators and Partner Organizations

1. Robert Read, Public Invention, PI
2. Victoria Jaqua, Public Invention, Co-PI
3. Avinash Baskaran, Public Invention, PostDoc

4. Andrew Lamb, Internet of Production Alliance, Letter of Collaboration
5. Sherry Lassiter, Fab Foundation, Letter of Collaboration
6. Ryan Fatica, Distributed Medical Device Manufacturing, Letter of Collaboration
7. Michael Jacobs, Distributed Medical Device Manufacturing, Letter of Collaboration
8. Julian Stirling, Open Flexure Microscope / GitBuilding, Letter of Collaboration
9. Jason Griffey, Open Science Hardware Foundation, Letter of Collaboration

List of Suggested Reviewers

Data Not Available

List of Reviewers Not to Include

Data Not Available